Gut–pulmonary vascular–right ventricular interactions in PAH

A Intestinal changes



Intestine (schematic)



Gut microbiota

Colonic lumen

Barrier impairment

Microbial compositional / functional shifts

Microbial function + host processing

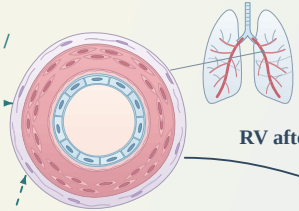
Candidate ecological response

Gut alterations

B Pulmonary vasculature

MAMP translocation / exposure

Altered metabolite exposure



Vascular remodeling

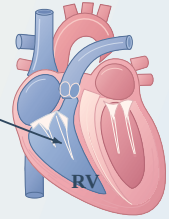
PVR ↑

RV afterload ↑

C Right ventricle

Stage dependent

Adaptation / maladaptive remodeling



With RV dysfunction

Systemic hemodynamic compromise

Venous congestion

Right atrial pressure ↑

Mesenteric hypoperfusion

Reduced mucosal perfusion

Gut wall edema

Human heart failure evidence [68–70]

Proposed feedback in PAH

L. rhamnosus GG	RV function improved	Pulmonary vascular load / pulmonary arteriolar remodeling
Experimental PH MCT rats [71]	RV fibrosis reduced	No significant differences detected

Microbiota-related signals

Established afterload relation

Hemodynamic effects

Proposed PAH link / incomplete pathway